School of Information Technology

Diploma in Applied AI & Analytic
IT3301 Applied Machine Learning
AY2026 Semester 1

Assignment Guide

30% of Module Grade Submission by 26th July 2026 at 2359 Hrs

1. **Introduction**
   This is an individual assignment and contributes to 30% of the ICA. The deadline for submission is Week 14, 26th July 2026, at 2359.

2. **Assignment Submission**
   The delivery of your assignment should be in two parts:
   a. Source codes files in **Google Colab** Jupyter notebook format. Please retain the output results during submission. Mark will be deducted without output.
   b. A recorded presentation in MP4 format with a maximum duration of 10 minutes. Present the key information from your jupyter notebook file/written report, with your face shown clearly.

   All the files in this assignment are to be submitted into PoliteMall by end of Week 14 - **26th July 2026 at 2359 Hrs**.

   Multiple submission attempts, if done before the submission deadline are allowed. Only the most recentsubmission will be graded.

   Late submission penalties will be applied in accordance as stated in Section 3.

3. **Penalty for Late Submission**
   a. Cap at 50% of base mark for submission within 5 calendar days from deadline
   b. From 6th day within deadline onwards, award 0 mark
   c. Note that "day" includes non-working days (Sat, Sun and public holidays).

Machine Learning assignment: A Machine Learning Approach to Predict Fraud

XYZ Cybersecurity is a startup that works on finding and stopping online threats. Their main goal is to build a strong machine learning model that can accurately detect fraud and other types of cyber attacks.

- Data Understanding and Preparation including feature engineering (3m)
- Model Selection (3m)
- Performance Measurement (3m)
- Hyperparameter Tuning (5m)
- Extra feature and consideration (3m)
- AI Ethic consideration (5m)
- Documentation and Code Quality (3m)
- Project Walk through (5m)

Marking Rubrics (Total 30m)

Category	Excellent (A)	Good (B)	Average (C)	Poor (D)
Data Understanding and Preparation including feature	Marks: 2.4 – 3.0 Conducts thorough and appropriate data understanding,	Marks: 1.6 – 2.3 Conducts appropriate data understanding and preparation	Marks: 0.8 – 1.5 Conducts some data understanding and preparation	Marks: 0.0 – 0.7 No data understanding and preparation.

engineering (3m)	preparation and applying appropriate data cleaning techniques.	but may overlook some aspects.	but may overlook important aspects or not apply appropriate techniques.	
Model Selection (3m)	Marks: 2.4 – 3.0 Simulate at least 4 models in your analysis. Chooses the most suitable machine learning model based on a deep understanding of the problem, data, and model characteristics.	Marks: 1.6 – 2.3 Simulate at least 3 models in your analysis. Demonstrates good understanding in choosing appropriate machine learning algorithms to achieve the model requirements.	Marks: 0.8 – 1.5 Simulate at least 2 models in your analysis. Demonstrates a basic grasp of selecting machine learning algorithms, though there may be gaps in considering all relevant factors.	Marks: 0.0 – 0.7 Simulate at least 1 model in your analysis. Shows some understanding of machine learning algorithm selection but lacks depth in considering important factors.
Performance Measurement (3m)	Marks: 2.4 – 3.0 Appropriate use and excellent interpretation of performance metrics for evaluating model performance.	Marks: 1.6 – 2.3 Appropriate use of performance metrics for evaluating model performance with minor misconceptions in interpretation.	Marks: 0.8 – 1.5 Use and interpretation of performance metrics with some misconceptions.	Marks: 0.0 – 0.7 Inappropriate use and interpretation of performance metrics.
Hyperparameter Tuning (5m)	Marks: 3.9 – 5.0 Carefully selects a comprehensive set of hyperparameters, considering the characteristics of the model and the dataset. Demonstrates an advanced understanding of the impact of hyperparameters on model performance.	Marks: 2.6 – 3.8 Chooses a good set of hyperparameters, taking into account the model and dataset characteristics. Shows understanding of the importance of hyperparameter tuning.	Marks: 1.3 – 2.5 Selects basic hyperparameters, but some choices may not be well-suited to the model or dataset.	Marks: 0.0 – 1.2 Fails to choose appropriate hyperparameters, resulting in poor model performance.

Extra features and considerations/recommendation (3m)	Marks: 2.4 – 3.0 Assessment of additional features or considerations /recommendation beyond basic requirements.	Marks: 1.6 – 2.3 Implementation of some additional features or considerations/recommendation beyond basic requirements.	Marks: 0.8 – 1.5 Basic implementation of extra features or considerations/recommendation, but lacking depth or innovation.	Marks: 0.0 – 0.7 Limited or no implementation of extra features or considerations/recommendations.
AI Ethic consideration (5m)	Marks: 3.9 – 5.0 Comprehensive and detailed documentation of ethical considerations, including proactive risk assessment. Transparent communication about every aspect of the AI system, including potential ethical challenges. Rigorous measures to mitigate biases, ensure fairness, and prioritize ethical considerations in decision-making.	Marks: 2.6 – 3.8 Comprehensive documentation of ethical considerations. Transparent communication about data sources, model decisions, and potential biases. Measures in place to address and mitigate biases and ensure fairness.	Marks: 1.3 – 2.5 Documentation of ethical considerations with an awareness of potential biases. Transparent communication about data sources and model decision-making. Consideration of fairness and potential impacts on different user groups.	Marks: 0.0 – 1.2 Basic documentation of ethical considerations. Limited transparency on data collection and model decision-making. Some acknowledgment of potential bias without concrete actions to address it.
Documentation and Code Quality (3m)	Marks: 2.4 – 3.0 Code is highly readable, well-structured and modular. Optimized for efficiency. Reveals a thorough and comprehensive understanding of the project, providing high- quality documentation that is clear, well-organized, and insightful.	Marks: 1.6 – 2.3 Code is generally readable and adheres to standard conventions, with adequate modularity. Demonstrates an understanding of the project through documentation, but there are areas that require improvement or may be lacking in certain aspects	Marks: 0.8 – 1.5 Code quality meets basic standards but may have minor issues in readability or modularity. Grasp the project adequately, and with documentation offering a fundamental overview but falling short in depth or detail in	Marks: 0.0 – 0.7 Code quality is significantly below standard, hindering comprehension, maintenance, and performance. The documentation is incomplete or unclear, making it challenging to grasp key aspects or severely lacks detail.

			specific areas.	
Video Presentation (5m)	Marks: 3.9 – 5.0 Provides a comprehensive code walkthrough, explaining the problem, methodology, and results clearly. The explanations cover key aspects of the code with depth and detail. Video with face shown clearly shown.	Marks: 2.6 – 3.8 Presents the code well, with some room for improvement in organization and clarity. The explanations cover key aspects of the code but may lack some depth or detail. Video with face shown clearly.	Marks: 1.3 – 2.5 The code explanation is satisfactory but may be lacking in organization and clarity. Explanations cover basic aspects of the code but lack majority of depth and detail. Video with face not shown.	Marks: 0.0 – 1.2 The code explanation is lacking in organization and clarity. Explanations are incomplete or unclear, making it challenging to understand key aspects of the code. No video.